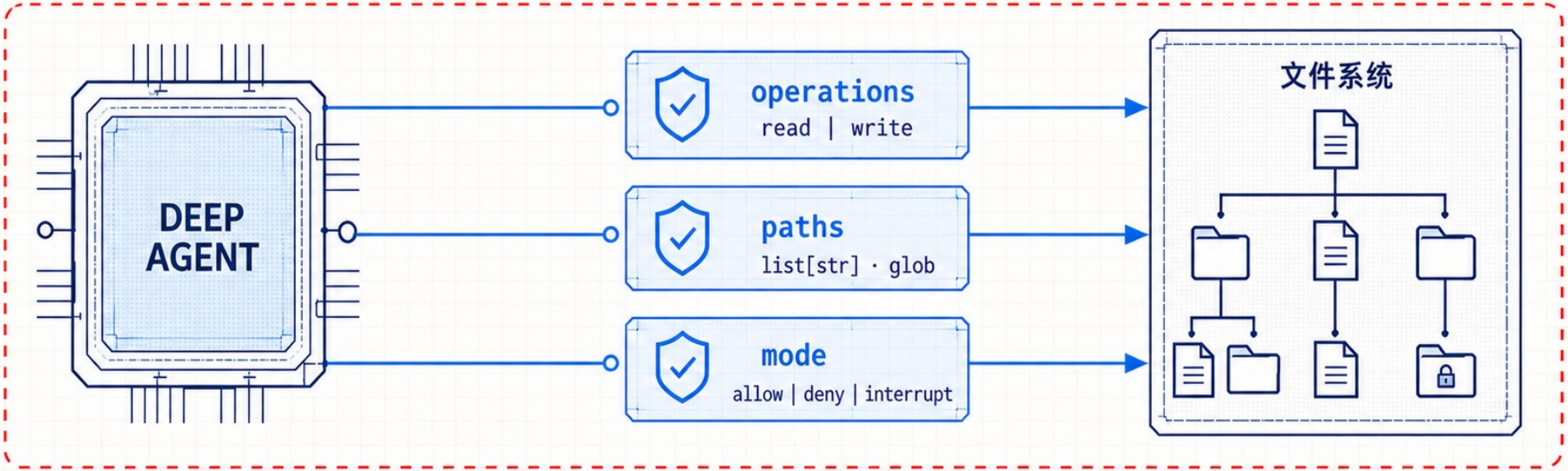
# Deep Agents 实战

## 第 15 讲：文件系统权限

FS.PERM.015
SCHEMA v1.0
BLUEPRINT

# 文件能力就是副作用入口

权限配置必须回答三个问题




**operations**

read → ls · read_file · glob · grep

write → write_file · edit_file · delete

**paths**

list[str] · glob

**mode**

allow | deny | interrupt

# 在入口函数中启用文件权限

`create_deep_agent(..., permissions=[...])`

```python
from deepagents import FilesystemPermission, create_deep_agent

agent = create_deep_agent(
    model=model,
    backend=backend,
    permissions=[
        FilesystemPermission(
            operations=["write"],
            paths=["/**"],
            mode="deny",
        )
    ],
)
```

## 这条规则如何生效

**operations=["write"]**

仅匹配写操作：write_file / edit_file / delete

**paths=["/**"]**

匹配所有绝对路径

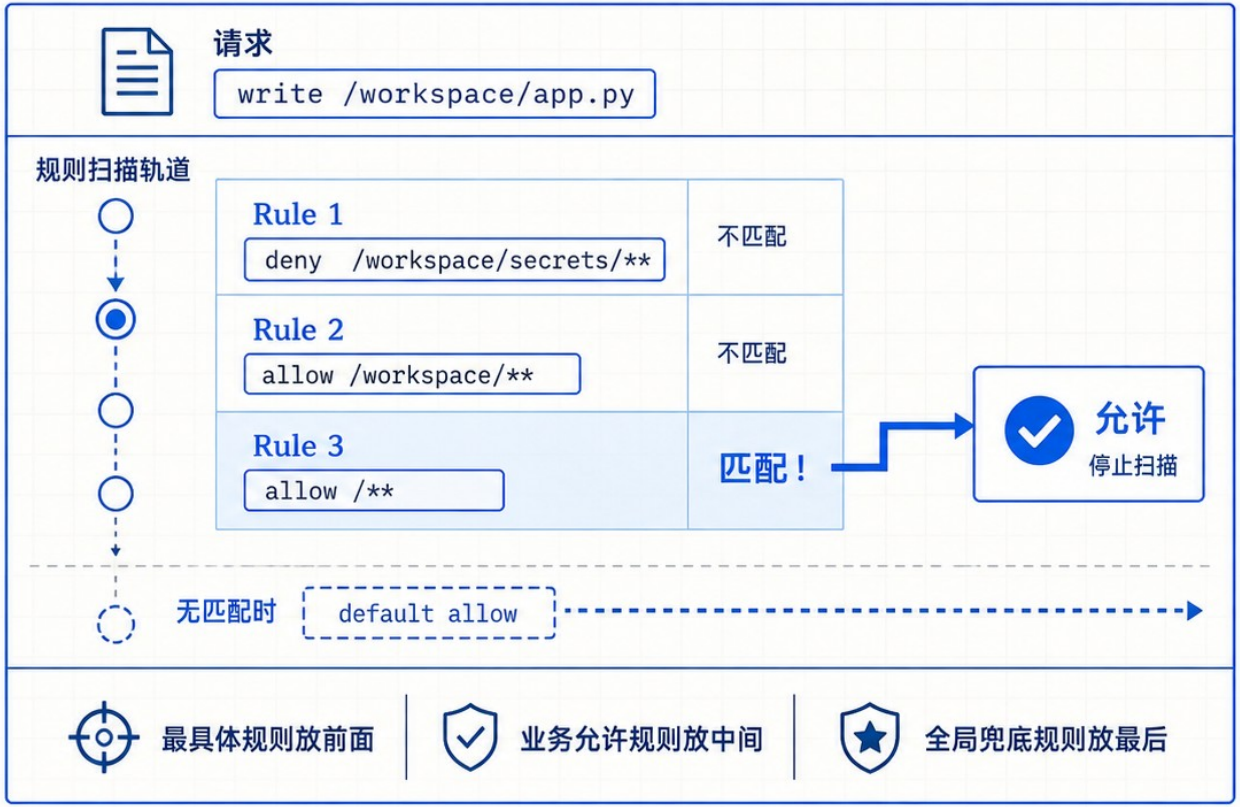
**mode="deny"**

命中规则后直接拒绝

**结果：所有写操作被拒绝；读取因未匹配此规则而默认允许**

# 规则顺序决定最终结果

first-match-wins；无匹配时默认允许





# 四种策略覆盖大多数场景

从最小副作用开始，再按职责放开

## ① 全局只读

`read: /**, write: none`

## ② 工作区白名单

/workspace

/**

`read: /workspace, write: /workspace`

## ③ 共享知识只读、用户记忆可写

/policies

/memories

`read: /policies, write: /memories`

## ④ 拒绝所有文件访问


`read: none, write: none`

# interrupt 把敏感写入交给人

暂停 → 审查 → approve / edit / reject → 恢复


需要 deepagents ≥ 0.6.8

需要 checkpointer

恢复必须复用同一 thread_id

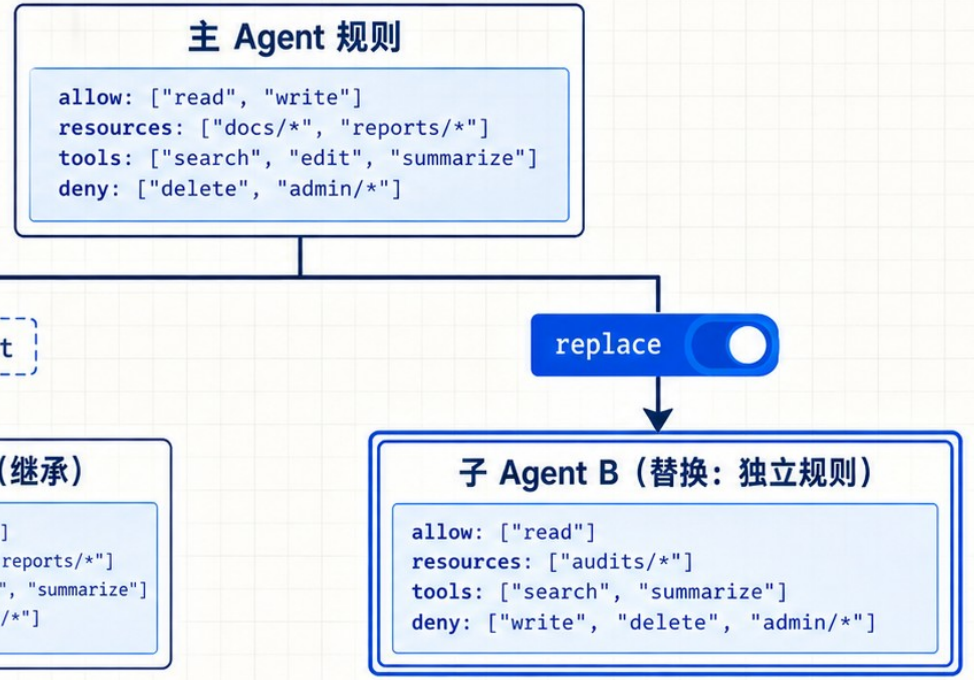
# 子 Agent 默认继承，显式配置整体替换

## 不是追加，也不是与父规则取交集

未配置：继承父 Agent 权限

已配置：完整替换父规则

子规则必须独立闭合




append

allow = parent.allow ∪ child.allow

resources = parent.resources ∪ child.resources

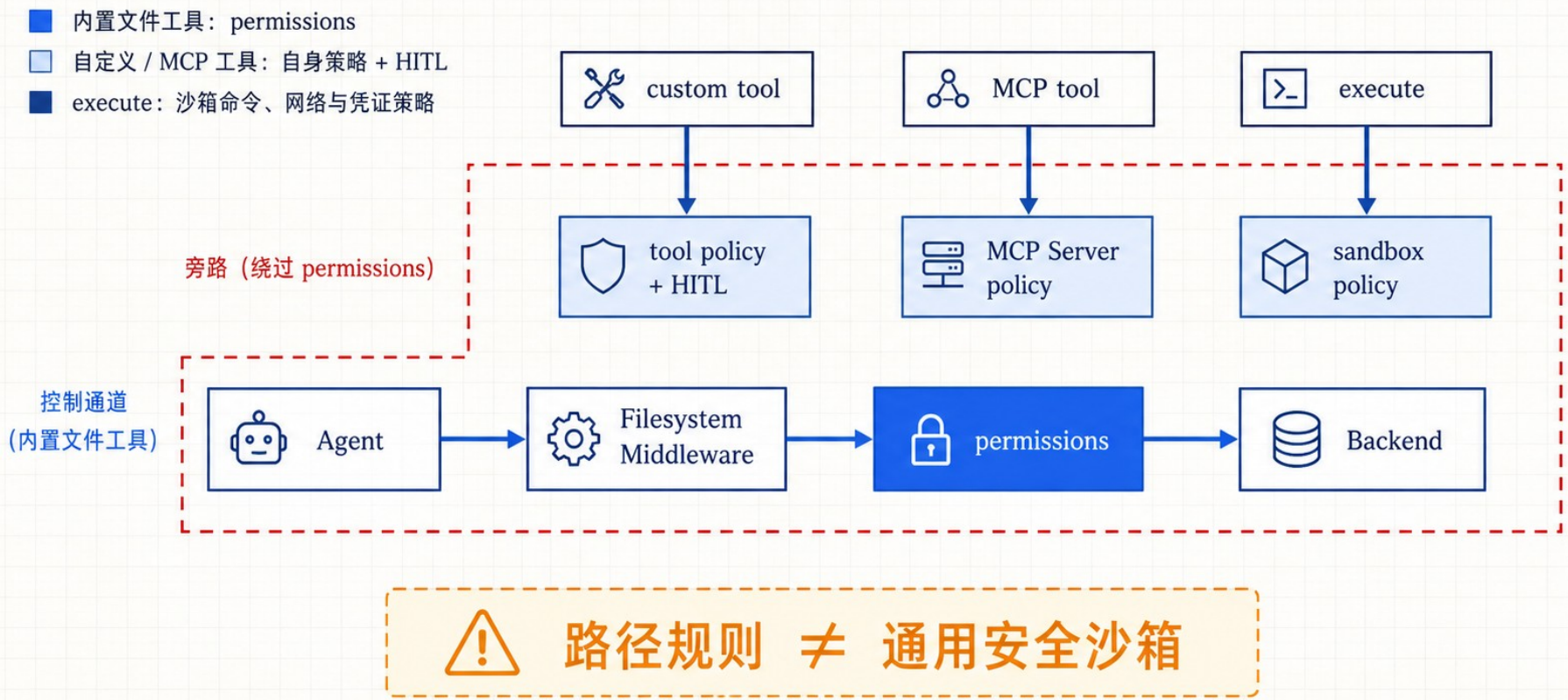
intersection

allow = parent.allow ∩ child.allow

resources = parent.resources ∩ child.resources

# 权限只能守住内置文件工具

custom tool、MCP 与 execute 都要单独设防



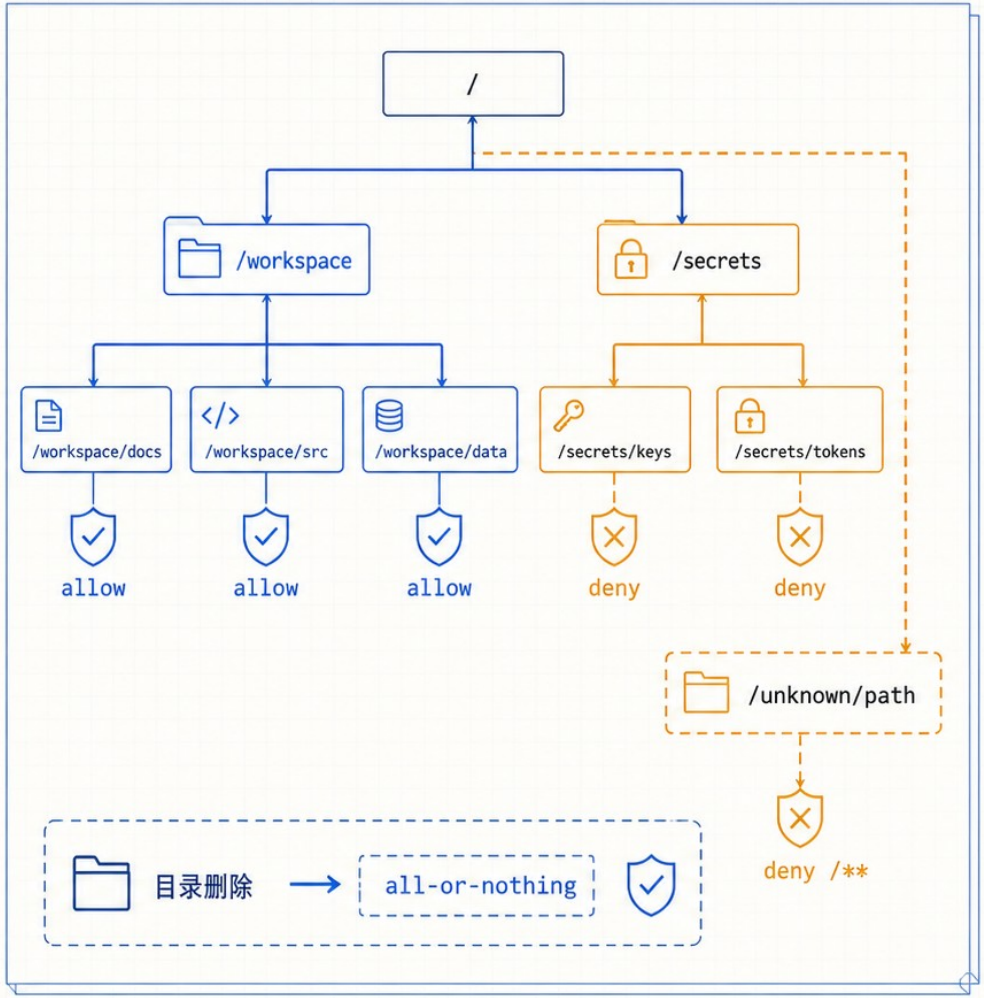

⚠ 路径规则 ≠ 通用安全沙箱

# 上线前验证的是安全保证

不仅要测允许路径，更要证明所有兜底边界

## 工程验收清单

- [x] 敏感路径：读 / 写 / 编辑 / 删除
- [x] 陌生路径：末尾兜底是否生效
- [x] 目录删除、子 Agent、interrupt、旁路



# 把最小权限写成可审查配置

稳定路径基线 · 少量人工审批 · 动态策略 · 沙箱隔离

- 先建立权限基线，再按 Agent 职责缩小边界